METRIC
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PERFORMANCE SPECIFICATION

WORLD GEODETIC SYSTEM 1984 (WGS 84)

Criteria for Spatial Reference Frame Alignment

This specification is approved for use by all Departments and Agencies of the Department of Defense.



Comments, suggestions, or questions on this document should be addressed to the National Geospatial-Intelligence Agency (NGA), ATTN: Departmental Standardization Officer, 3838 Vogel Road, Arnold, MO, 63010-6238, Mail Stop L66, or by e-mail to DepSO@nga.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1 SCOPE

1.1 Scope

This standard establishes the requirements and specifications for the World Geodetic System 1984 (WGS 84) Reference Frame (RF) alignment to the International Earth Rotation and Reference Systems Service (IERS) International Terrestrial Reference Frame (ITRF) by the National Geospatial-Intelligence Agency (NGA) Office of Geomatics, Global Navigation Satellite System (GNSS) Division.

1.2 Purpose

The purpose of this document is to outline the alignment of the WGS 84 RF, note the variables that impact the alignment process, and identify triggering mechanisms for additional work.

2 APPLICABLE DOCUMENTS

2.1 General

The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government Documents

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.2.1.1 Department of Defense Standards

- MIL-STD-961E, Defense and Program-Unique Specifications Format and Content - 16 Jul 20 (https://quicksearch.dla.mil/qsDocDetails.aspx?ident number=36063)
- MIL-STD-962D, Defense Standards Format and Content - 9 Jan 14

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(https://quicksearch.dla.mil/qsDocDetails.aspx?ident number=36064)

- CJCSI 3900.01D, Position (Point and Area) Reference Procedures (https://www.jcs.mil/Library/CJCS-Instructions/)
- CJCSI 6130.01G, Master Positioning, Navigation, and Timing Plan (Limited Distribution: Email DepSO@nga.mil for assistance)
- Department of Defense Directive 4650.05, Positioning, Timing, and Navigation (PNT) (https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/465005p.pdf).

2.2.1.2 National Geospatial-Intelligence Agency Standards

- NGA.STND.0036_1.0.0_WGS84, Department of Defense World Geodetic System 1984: Its Definition and Relationships with Local Geodetic Systems, Version 1.0.0, 8 July 2014 (https://nsgreg.nga.mil/registries/search/results.jsp?registryType=doc&searchTextField=NGA.STND.0036_1.0.0&searchOnField=name&registerField=any&itemTypeField=doc&itemTypeField=dst&statusNameField=2&nsgKeywordsField=any&disrServiceAreaField=any&stdStatusField=any&dayField=4&monthField=1&yearField=2023&tzOffset=360&searchButton=)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

- National Space Policy 12/9/2020 (https://www.space.commerce.gov/policy/national-space-policy/)
- 2019 Federal Radionavigation Plan (https://rosap.ntl.bts.gov/view/dot/43623)

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- NAVSTAR GPS Control Segment/National Geospatial-Intelligence Agency Interfaces ICD-GPS-211 Rev. H. Email DepSO@nga.mil for assistance.
- NAVSTAR GPS Control Segment/National Geospatial-Intelligence Agency Interfaces ICD-GPS-811 Rev. F. Email DepSO@nga.mil for assistance.

2.3 Non-Government Publications

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

- International Earth Rotation and Reference Systems Service (IERS) Conventions 2010 - Technical Note No. 36 (https://www.iers.org/IERS/EN/Publications/TechnicalNotes/tn36.html)
- Agreement on the Promotion, Provision and Use of Galileo and GPS Satellite-Based Navigation Systems and Related Applications (https://safe.menlosecurity.com/doc/docview/viewer/docNDA5EBC3A85E050b6dd795e132984dd657e5445d69c8feec70cefb2fb6ffd83cd2f77384b83de)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of Precedence

Unless otherwise noted herein, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 General Description

NGA's GNSS Division is responsible for the development and maintenance of the WGS 84 Reference Frame. That reference frame is realized through the set of Department of Defense (DoD) ground station positions and data used to monitor the GPS (Global Positioning System) constellations. The NGA GPS Precise Ephemeris (PE) serves as the DoD truth for the WGS 84 location and on-board clock states for the active GPS

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satellites. Its use is mandated for two important DoD/Intelligence Community (IC) functions (CJCSI 6130.01G, Enclosure A, section 3.h.):

- For the GPS Operational Control Segment (OCS) to monitor the accuracy of their Kalman Filter's estimates of the same quantities.
- As the PE used in GPS Precise Point Positioning (PPP), resulting in PPP solutions based on WGS 84.

NGA's GNSS Division is responsible for periodically updating the alignment of WGS 84 to international standards. Until this document, there has been no requirement as to how often or accurate the WGS 84 alignment must be, only according to CJCSI 6130.01F that: "The WGS 84 is consistent within its precision with the International Terrestrial Reference Frame maintained as the international standard by the International Earth Rotation Service. NGA ensures that WGS 84 is coincident with the International Terrestrial Reference System which contributes to GPS interoperability with foreign GNSS."

By both policy and practice, the WGS 84 terrestrial reference frame as realized by the WGS 84 reference stations is aligned as closely as possible to the ITRF. Due to the use of multiple techniques, a large number of well-distributed ITRF reference stations and highly rigorous estimation methods, the ITRF is considered the best available Terrestrial Reference Frame (TRF). In turn, the International GNSS Service (IGS) produces a frame from a subset of well-performing IGS stations achieved specifically via Global Navigation Satellite System (GNSS) data. It is the current realization of the IGS frame to which we align WGS 84.

3.2 Units of Measure

The realization of the WGS 84 Reference Frame comes from a set of positions and velocities of a set of fixed stations. To keep the frame accurate over time, these locations are also assigned velocities to keep the frame aligned to the underlying movements of the Earth's crust. As such, the coordinates that define the frame are given in Cartesian coordinates in an Earth-Centered, Earth-Fixed (ECEF) coordinate system with units of meters (m) and the corresponding velocities are given in meters per year (m/yr).

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When comparing reference frames, it is common to use a 7-parameter Helmert transformation. The seven parameters are displacement in each of the X, Y, and Z directions, rotations about each of the X, Y, and Z-axes, and a scale factor. Displacements are measured in meters (m), rotations are measured in milli-arcseconds (mas), and scale factor is in parts per billion (ppb).

3.3 Sign Conventions

The WGS 84 Coordinate System is a Conventional Terrestrial Reference System (CTRS). The definition of this coordinate system follows the criteria outlined in IERS Technical Note No. 36, IERS Conventions 2010, Section 4.1 (IERS TN 36). These criteria are repeated below:

- It is geocentric, the center of mass being defined for the whole Earth including oceans and atmosphere
- Its scale is that of the local Earth frame, in the meaning of a relativistic theory of gravitation
- Its orientation was initially given by the Bureau International de l'Heure (BIH) orientation of 1984.0
- Its time evolution in orientation will create no residual global rotation with regards to the crust

The WGS 84 Coordinate System is a right-handed, Earth-fixed orthogonal coordinate system and is graphically depicted in Figure 3.1.

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IERS Reference Pole (IRP)

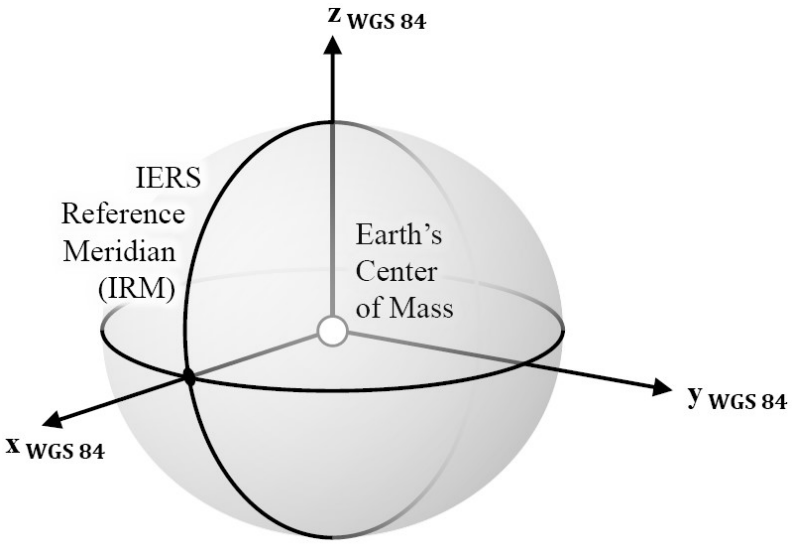


Figure 3.1 The WGS 84 coordinate system definition

In Figure 3.1, the origin and axes are defined as follows:

- Origin = Earth's center of mass
- Z-Axis = The direction of the IERS Reference Pole (IRP). This direction corresponds to the direction of the BIH Conventional Terrestrial Pole (CTP) (epoch 1984.0) with an uncertainty of 0.005 inches
- X-Axis = Intersection of the IERS Reference Meridian (IRM) and the plane passing through the origin and normal to the Z-axis. The IRM is coincident with the BIH Zero Meridian (epoch 1984.0) with an uncertainty of 0.005 inches
- Y-Axis = Completes a right-handed, Earth-Centered Earth-Fixed (ECEF) orthogonal coordinate system

The WGS 84 Coordinate System origin also serves as the geometric center of the WGS 84 Ellipsoid, and the Z-axis serves as the rotational axis of this ellipsoid of revolution.

The definition of the WGS 84 CTRS has not changed in any fundamental way since its original implementation. This CTRS continues to be defined as a right-handed, orthogonal, and Earth-fixed coordinate system which is intended to be as closely coincident as possible with the CTRS defined by the IERS or, prior to 1988, its predecessor, the BIH.

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3.4 Definition of Accuracy

For the purposes of this Performance Specification document, the definition of accuracy for the WGS 84 Reference Frame will be its alignment to the internationally accepted standard for all global spatial reference frames, i.e. the ITRF realization produced by the IGS.

3.5 Alignment Criteria

The WGS 84-to-ITRF alignment metrics are:

1. 7-Parameter Helmert Transformations between the NGA-to-IGS GPS Precise Ephemerides.
   a. A 7-Parameter Helmert Transformation is a standard frame alignment metric used in geodesy. Since GPS Precise Orbits are a reflection of their underlying terrestrial frame, a 7-parameter orbit comparison between two sets of PEs is also appropriate for frame alignment measures.
      The seven parameters are:
      - Translation in the X, Y, and Z directions
      - Rotation along the X, Y, and Z axes
      - Scale difference between the two PEs
   b. The differences in these frames are computed on the surface of the earth, using a meant earth radius, then a root sum square (RSS) of those seven can serve as a single statistic.

2. PPP solutions, using NGA GPS orbit/clock, of ITRF sites and the differences with their published coordinates.
   a. Ultimately, the GPS orbit/clock data is used for geolocation. Using the NGA GPS Precise Ephemerides to compute WGS 84 coordinates for established sites with ITRF coordinates are alignment metrics. NGA compares derived-vs-published IGS site coordinate differences as a tool to monitor the WGS 84-to-ITRF alignment.

3. PPP solutions, using IGS GPS orbit/clock, of NGA sites and the differences with their IGS SINEX (Station Independent Exchange) coordinates.
   a. NGA submits a subset of daily station data to the IGS. In return, NGA's station coordinates in the ITRF are available in the IGS SINEX solutions. A

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time series of those IGS coordinates are constructed and compared to internally created data sets of the same station. These compares serve as a consistency check between WGS 84 RF and ITRF. Should the Root Mean Square (RMS) of the total magnitude differences over all stations during the previous six months exceed 2 centimeters, NGA will begin a new WGS 84 RF-to-ITRF alignment.

Should a new ITRF be released or if any of the following exceed the experience-based tolerance metric of 2 centimeters over a six-month period, the Office of Geomatics in the NGA will evaluate the need for a new WGS 84 RF alignment to the current ITRF:

1. the RSS of the 7-parameter transformation or,
2. the East, North, Up component RSS of the PPP solutions or,
3. the NGA-vs-SINEX comparisons

3.6 Product Delivery

Delivery of the WGS 84 Reference Frame occurs when NGA:

1. Provides new coordinates for the tracking antenna at the NGA/United States Space Force (USSF) GPS Monitoring Network to the United States Space Force, 2nd Space Operations Squadron. Details of this product delivery are outlined in ICD-GPS-211, Architecture Evolution Plan (AEP) and ICD-GPS-811, Next Operational Control Segment (OCX).

2. Uses the new GPS Monitoring Station coordinates at NGA's Monitor Station Network Control Center (MSNCC) software and GPS Orbit Determination process.

Once the new coordinates are implemented in the USSF GPS Operations Control Segment, users of GPS for positioning will automatically be utilizing the new WGS 84 Reference Frame. Additionally, when the new coordinates have been implemented in NGA's MSNCC and GPS Orbit Determination software, NGA's GPS precise ephemeris will also be in the new reference frame. Notifications will be sent as appropriate.

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3.7 Security Classification

This document in its entirety is Unclassified. The WGS 84 is an Unclassified product of the NGA.

4 VERIFICATION

4.1 Conformance Inspection

The process implemented provides a regular inspection of conformance/variance to accepted accuracy specifications up to but not exclusively a daily cadence.

4.2 Technical Review

NGA's subject matter experts will review the scientific efforts of the producer during the preparation of this performance standard document.

5 PACKAGING

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

5.1 Packaging

When packaging of material is to be performed by DoD or in-house contract personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, media products, or by contacting the responsible packaging activity.

6 NOTES

6.1 Intended Use

This specification is intended to provide guidelines for the calculation of the WGS 84 alignment regarding process, accuracy, and frequency of alignment calculation.

The WGS 84 is considered Public Release. This specification may also be used by public, private, academic, scientific, and other domains for the application of WGS 84.

6.2 WGS 84 Alignment Data

The data used to in the alignment calculations for the WGS 84 Reference Frame is observational tracking data from established GPS tracking stations with known positions.

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6.3 Acquisition Requirements

Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of the DISR to be cited in the solicitation and, if required, the specific issue of individual documents referenced.

c. When a first article is required (see 3.1 and 4.2).

d. Levels of packaging (see 5.1).

6.4 Process Description

6.4.1 Background

By both policy and practice, the WGS 84 TRF as realized by the WGS 84 reference stations is aligned as closely as possible to the ITRF. Due to the use of multiple techniques, a large number of well-distributed ITRF reference stations and highly rigorous estimation methods, the ITRF is considered the best available TRF. In turn, the IGS produces a frame from a subset of well-performing IGS stations achieved specifically via GNSS data. It is the current realization of the IGS frame to which WGS 84 is aligned. Constructing the WGS 84 frame that is independent but aligned with the ITRF/IGS frame provides two advantages: 1) the frame is composed of stations that are under DoD control, and 2) the frame takes advantage of the best-available international scientific standard. Over time, the WGS 84 and ITRF/IGS frames may diverge owing to unmodeled motions of the stations that define the respective frames. When this occurs, alignment between WGS 84 TRF to the ITRF/IGS TRF is needed (see 3.5).

When NGA makes an alignment of the WGS 84 Reference Frame, it is important to align it as closely as possible to the ITRF for two reasons. First, ITRF is tied to the celestial sphere by stations on Earth that observe known celestial markers (typically a quasar). This is how it is known how much the Earth wobbles and the Earth’s rate of rotation. It also gives a fixed reference frame despite all plate tectonic motion occurring on the Earth’s surface. If one referenced to markers on the Earth’s surface only, that reference frame would be constantly moving, depending on which tectonic plate one is on. The celestial sphere, even though it is slowly

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moving over eons, is the closest thing to a stationary reference. Second, NGA should match as closely to ITRF since ITRF is the reference frame used by the GPS community outside of DoD. Process

The alignment for the WGS 84 Reference Frame historically was constructed from GPS satellite measurement data from NGA and USSF tracking stations and are supplemented with data from the global network of various sites operated by a wide variety of public and private agencies of domestic and foreign origin which adhere to the scientific standards and nomenclatures adopted by the International GNSS System.

However, a new method is now used to produce an updated alignment of the WGS 84 Reference Frame to the ITRF. This method utilizes a multi-year time history of the current NGA/ USSF GPS monitoring stations in the ITRF. A least squares solution is applied to determine both the position and velocity of the station. The method produces position results that agree with the international accepted “truth” data, i.e., positions of these stations as calculated by the IGS in the low-mid millimeters per station per component. This method not only produces more accurate and more precise results than the previous network solution technique but does while requiring less labor and computing resources.

6.5 Subject term (keyword) listing

a. Key Words
alignment
center of mass
coordinate system
ellipsoid
reference frame
transformation

b. Acronyms

AEP	Architecture Evolution Plan
BIH	Bureau International de l’Heure
CJCSI	Chairman Joints Chiefs of Staff
CTP	Conventional Terrestrial Pole
CTRS	Conventional Terrestrial Reference Frame
DoD	Department of Defense
ECEF	Earth-Centered, Earth-Fixed
GNSS	Global Navigation Satellite System
GPS	Global Positioning System

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IC	Intelligence Community
ICD	Interface Control Document
IERS	International Earth Rotation and Reference Systems Service
IGS	International GNSS System
IRM	IERS Reference Meridian
IRP	IER Reference Pole
ITRF	International Terrestrial Reference Frame
MSNCC	Monitor Station Network Control Center
NGA	National Geospatial-Intelligence Agency
OCS	Operational Control Segment
OCX	Next Operational Control Segment
PE	Precise Ephemeris
PPP	Precise Point Positioning
RF	Reference Frame
RSS	Root Sum Square
SINEX	Station Independent Exchange
TRF	Terrestrial Reference Frame
USSF	United States Space Force
WGS 84	World Geodetic System 1984

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CONCLUDING MATERIAL

Custodians:
NGA – MP
Army – AV
Navy – NO
Air Force – 09

Preparing Activity:
NGA - MP
Project GINT-2022-001

Review Activities:
Army – MI
Navy – CG, MC
Air Force – 33, 99
DIA – DI
DISA – DC1
NSA – NS

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